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Non-linear finite element model established on pectoralis major muscle to investigate large breast motions of senior women for bra design

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Abstract

This paper presents a three-dimensional nonlinear biomechanical finite element model to simulate elderly breast deformation under arm abduction. The finite element model was constructed based on complex anatomical structures that consist of the soft tissues including torso, breast, pectoralis major muscle, and rigid bones including humerus, sternum, clavicle and ribs. The elderly breast was defined as material nonlinearity and geometric nonlinearity. The computational model can simulate and illustrate the deformation of soft tissues during arm abduction from 30° to 90°. The finite element model was validated by motion data. The Young's modulus for the clavicular portion and the sternocostal portion of the pectoralis major muscle was characterized as 0.1 MPa and 0.08 MPa, respectively. Besides this, the finite element model presented here features the musculoskeletal system for breast deformation and reveals the synergistic relationship between the breast and the pectoralis major muscle. A questionnaire was conducted to analyze the importance of purchasing factors and the discomfort positions in a sports bra from the perspective of senior women. Combined with the finite element model results, this study provides a promising basis for the design of sports bras in an ergonomic way.

Keywords

Finite element model, breast, pectoralis major muscle, motion analysis, senior women, bra design

The demographic pressure of the aging population is one of the biggest concerns that will place an unprecedented increase in public expenditures and financial portfolios of individual families.[1,2] The elderly are encouraged to be more active in some low-intensive activities, which is beneficial for decreasing detrimental consequences imposed on the social healthcare system.[3] Yoga or Taichi are highly recommended under the consideration of special physiological and psychological exercise capabilities of the elderly.[4] However, as earlier studied that exercise-induced breast pain or discomfort significantly hindered women in pursuing this activity.[5] Excessive skin strain of breasts generated from exercise or ineffective external/internal support may lead to certain breast problems.[6] For example, typical postures in a yoga practice such as the tree pose and warrior I usually require practitioners to stretch their chests to a large extent. In this situation, soft tissues of the breast are undergoing large deformations.

It was reported that the mean failure strain of exercised human skin samples was $54\% \pm 17\%$ at a strain rate of $0.012\,s^{-1}$.[7] Under the bare breast condition, skin strains can reach up to 93% during running.[8] Far less is known of the bare breast strain under some stretching

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low-intensive activities. To have a better understanding of the potential skin strain magnitude, this study will detect the elderly breast deformations under arm abduction to avoid unnecessary breast risks.

To measure the skin strain of the breasts, photosonics cameras or infrared marker-tracking cameras were used in the traditional technologies. Usually, the center of the breast (nipple or a little bit higher than the nipple) was selected as the representative point to present the whole breast motion,[9] but the abstracted information can not accurately describe the local breast deformation. Besides, most previous studies focused on characterizing the breast-torso movements to analyze how the intensive breast bounce influences the perceived breast discomfort for better high-intensive sports bra design. However, as for senior women, studying the global and detailed information on breast deformation during moderate and stretching activities is of significant importance.

Recently, considerable progress has been made on the finite element models (FEMs) of breast deformation in recent years.[10,11] Li et al. built a three-dimensional (3D) biomechanical FEM consisting of elastic breast, rigid body, and bra to simulate the bra–breast interaction.[12] Liang et al. simulated the static and dynamic contact FEM that consists of a body and sports bra.[10] Sun et al. constructed FEM including a rigid torso, deformable breast, and bra without underwire.[11] However, these simplified geometric components involved in the FEM cannot be accurate enough to observe global breast mechanics.

Therefore several concerns need to be addressed. First, the majority of studies focused on young women, limited studies focused on senior women. The anatomical structure and corresponding mechanical properties for tissues in breasts vary significantly between the young and the elderly. Senior women should be paid more attention. Besides this, walking and jogging were the mainstream research. Other exercises that consist of relative static and large body stretching were neglected. Finally, the traditional strategy was to establish FEM that was directly built based on breasts for investigation of their deformation.[13]

The anatomical structure of a normal female breast is shown in Figure 1. It overlays the pectoralis major muscle (PMM), which covers from the second rib to the sixth rib.[14] Without muscles inside the breast, the intrinsic structural support tissues that provide major anatomical support are Cooper's ligament and skin.[15,16] The breast deformation is believed to be as a result of whole musculoskeletal mechanics, rather than solely depending on the breast tissues. It might be driven by the static/dynamic shear forces, compression/tension on the pectoralis muscles, or arm motions.[17] Besides, the constitutive model of the soft tissues also plays an important role in influencing breast deformation. Different from younger women, the elderly breast shows a significantly different structure and morphology due to the hormonal variability during different life stages, such as the menstrual cycle, breast-feeding period, menopause, and aging.[18] Influenced by the aging process, reproductive hormone estrogen declines, skin, and other connective soft tissues become less elastic, and the breast tissues become less dense. The breasts will be dominated by fatty tissues around 55 years of age.[19] Hence, for senior

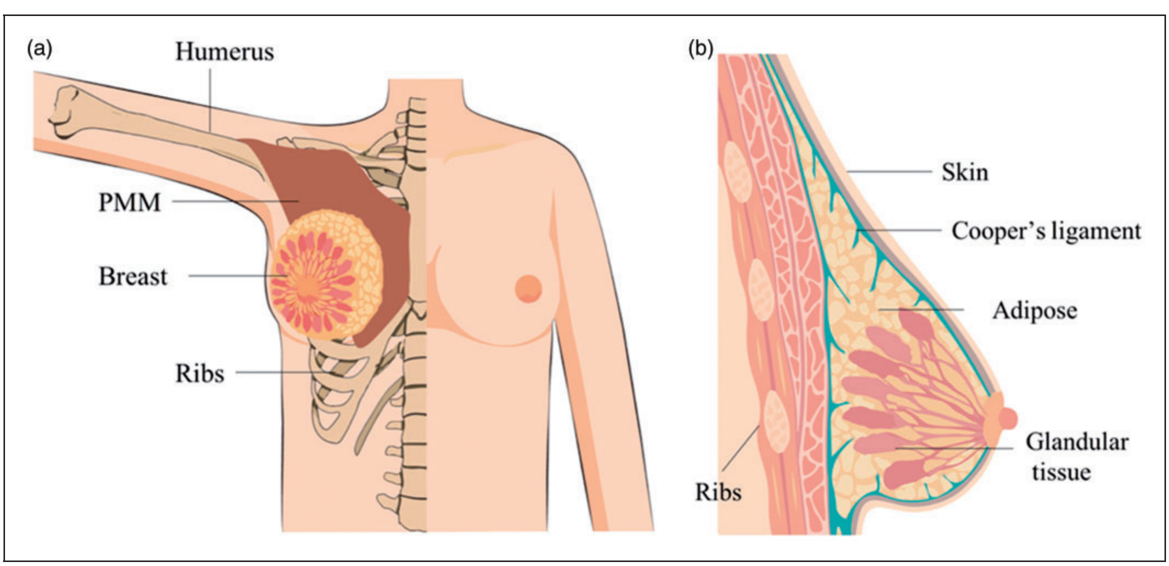

Figure 1. Musculoskeletal system with breast structures. (a) A simplified anatomical diagram of the female torso; (b) sagittal section of the breast.

women, the breast can be simplified to be assumed as a homogenous structure in some cases. Due to the different anatomical structures and material properties, it is inaccurate to apply the results directly derived from younger women to senior women.

In this study, a survey was conducted to explore the purchasing factors and discomfort positions of sports bras from the perspectives of senior women. For achieving relatively large and safe chest stretching,[20] arm abduction was selected. Based on the collected breast motion data, an anatomical-based FEM was built. The FEM consisted of the soft torso, soft breast, deformable PMM, and rigid skeletons. The FEM aims to study the deformation of soft tissues in senior women's torsos during arm abduction. A new insight posed here will add to having a better understanding of breast motions during exercise, which may inspire the fascinating industrial bra design to expand the commercial market for women, especially senior women.

Methods

Experimental works

Questionnaire. In order to evaluate the bra design, a survey about the purchasing factors and preferences for the sports bra was carried out online (wj.qq.com; Tencent Holdings Limited) among Chinese women. In total, 328 results were obtained with 69.8% women aged below 39 years and 30.2% aged above 40 years. Analogous to previous surveys, some variables such as participants' demographics, purchasing behavior and the discomfort of their sports bras were included. Some personal information such as age was used as options (e.g. 60–65 years). For the investigation of the importance of purchasing factors, a 7-point Likert scale was selected to assess the purchasing factors from the perspective of senior women. The 1 to 7 points correspond to strongly disagree, disagree, more or less disagree, undecided, more or less agree, agree and strongly agree. The detailed question items in the questionnaire is shown in Table 1. Each categorization was divided into six sub-items.

Motion data collection. Breast deformation was collected and characterized first, and then used for the FEM validation. A motion capture system (Eagle, USA), 12 high-resolution digital cameras, was adopted to track and record trajectories of those small spherical retro-reflective markers (diameter is 9.5 mm, mass is 1.81 g). The captured frame was illustrated (Figure 2(c)).

As sagging is one of the most obvious signs of the elderly breast, its inframammary fold is uncovered.

Table 1. The detailed question items to investigate the purchasing factors of sports bras

Categorization	Items
Quality	Textile quality/sewing quality/accessories' quality/coloring quality/making and labeling/measurement specifications
Functions	Support/stable control/skin covering/easy to take on and off/bouncing control/not move up and down
Comfort	Suitable sizing/permeability/suitable pressure/sweat breathe ability/fit/elasticity
Aesthetics	Breast uplift/breast gathering/color/outlook design/design style/beauty
Recommendations	Friends/advertisement/branding/sales promotion/health promotion/confirmity

The focus of this study is to measure the bare breast condition without external bra support. It is difficult to capture the bottom area of the sagging breast, hence only the upper area of the breast is included in this study. Figure 2(a) shows the position of retro-reflective markers that were directly attached on the nude breast. To be specific, the nipple was marked as O and marker B was 11 cm above O in the vertical direction. According to the breast boundary measurement, once OB was identified, we can locate the MN that was perpendicular to OB. Precisely, M was in the middle line of the body and N is in the breast root. For the other two markers, A and B, they could be traced based on the following information: $\angle AOB = 75°$, $\angle COB = 35°$ and $OA = OC = 11$ cm. Before data recording, a calibration was done for the motion capture system. The participant was kept standing in the assigned place for camera shooting, followed by her arms steadily abducting from 30° to 90° (Figure 3(b)). Because OB is the central line of the breast, the opposite OA and OC pulls OB to deform, the strain of OB largely depends on OA and OC. Only the strain of OA and OC was measured to analyze the breast deformation. In this process, the real-time 3D coordinates corresponding to markers A and C were measured and recorded by the motion capture system.

Large breast size was reported to have numerous negative health problems, such as back and neck pain, which are key factors to prevent women from being physically active. Bra–breast forces generated in women with large breasts while standing and during treadmill running have implications for sports bra design. Hence, those participants with a history of breastfeeding and no breast surgery history, with a cup size above C were recruited.

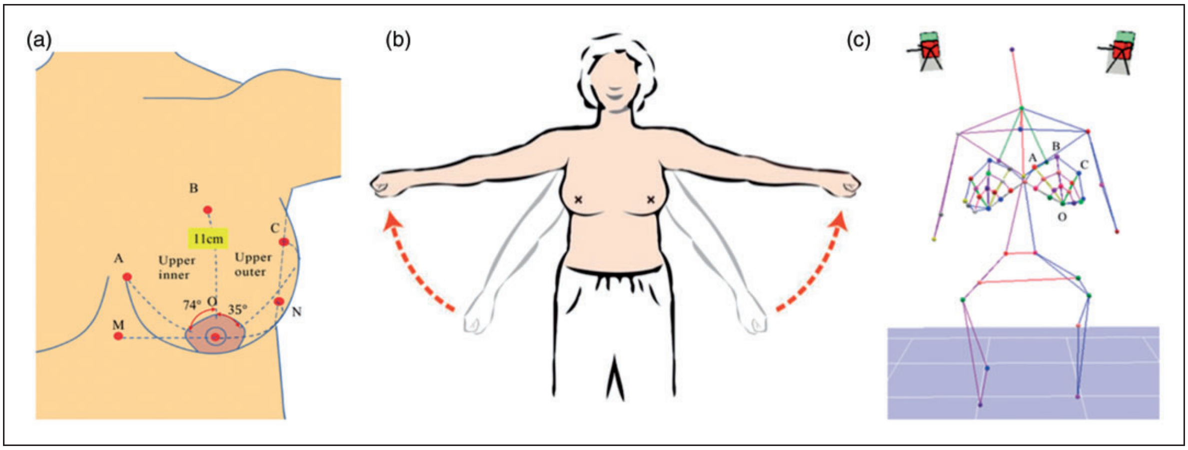

Figure 2. Experimental design and manipulation. (a) The detailed information of each marker; (b) arm abduction from 30° to 90°; (c) typical frame obtained by the motion capture system.

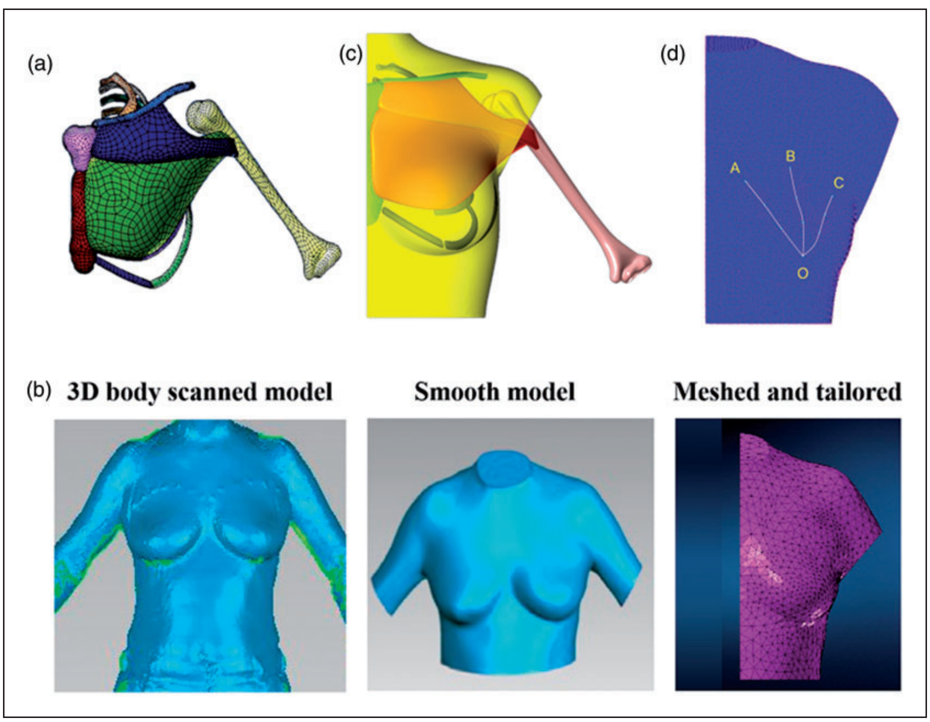

Figure 3. The construction of a geometric model based on musculoskeletal systems. (a) The musculoskeletal structure; (b) the 3D geometric model generated by 3D body scanner, processed geometric model and meshed and tailed geometric model; (c) the boundary condition for the finite element model (FEM); (d) the location of markers A, B, C and O on the FEM.

Six healthy female volunteers were invited for the experiments with an average age of 61 years, height of 155 cm, weight of 56.1 kg, underbust girth of 79 cm, breast circumference of 91 cm and cup size of 80C, as detailed in Table 2. Informed consents were obtained from participants with regard to the whole experimental process including 3D scanning and motion capturing of nude breasts. All experiments were performed in compliance with the guidelines of the human subjects ethics sub-committee and approved by the ethics committee at The Hong Kong Polytechnic University. The data from participant no. 5 were selected for constructing the FEM.

FEM construction

Individual-specific 3D body geometric model. In order to improve the computing efficiency and accuracy, the FEM was simplified under the assumptions that the breast lies on the specific PMM without considering

Table 2. The detailed body information of participants

Participant no.	Age (years)	Height (cm)	Weight (kg)	Cup size	Underbust girth (cm)	Breast circumference (cm)
1	58	159	53.5	70C	72.1	80.3
2	66	154	54.2	75C	75.6	91.0
3	57	147	57.0	85C	82.0	94.7
4	60	160	62.1	90C	88.7	102.6
5	66	148	53.1	80C	79.3	90.0
6	61	159	56.7	80C	77.0	89.6

joints and muscle fascia. The elderly breast was assumed as fat tissues only. Humerus, clavicle, the first to the sixth rib, sternum and PMM were used in this study. Arm abduction, which is the most common posture with large chest stretching was adopted.[19]

The 3D geometric model of musculoskeletal system was acquired from @TurboSquid (TurboSquid, USA), as shown in Figure 3(a).

The 3D body scanner (VITUS Smart XXL, Germany) was used to get an individual-specific geometric model (Figure 3(b)). It is worth noting that the participant was required to put on a standard bra (Wacoal, Wacoal H.K. Co., Ltd.) to visualize breast roots because of the sagging problem. In order to achieve a smooth and cutting-out geometric model capable of constructing the FEM, post-treatment has been done on the original geometric model by using three softwares. First, the model was processed by Geomagic Design X (Rapidform XOR software, USA) to fill pores on the breast's surface, smooth uneven surfaces and delete reduplicated layers. Second, Rapidform (Rapidform XOR software, USA) was employed for faces subdividing, shape trimming and auto-surfacing for the following mesh creation. Third, as reported, the PMM could be divided into two heads.[14] One is the claviclular head covering half of the clavicle, while the other is the sternocostal head located from the second to the sixth ribs and costal margin of the sternum. The clavicular and sternocostal portion were applied in this work for the PMM by Apex (MSC, USA).

Mechanical properties. All of the breast, torso and PMM can be regarded as isotropic, quasi-incompressible and nonlinear materials.[21] A number of constitutive material models have been established for different muscles and soft tissues. As reported by our group,[22] the Moony–Rivlin coefficients of the breast were determined specifically for the elderly in a body forward leaning process. Here we extend the application of these material coefficients in an arm abduction. In general, the skin thickness varies from 521 to 1977 μm through the whole body. While for the breast skin of elderly women, a thickness ranging from 1 mm to 2 mm has been described elsewhere.[23] Meanwhile, the elastic modulus of breast skin was calculated as 0.2~3 MPa. Therefore, in this model, the Young's modulus was set as 0.5 MPa and the Poisson's ratio was set as 0.49 due to quasi-incompressible characteristics of skin. Not surprisingly, the modulus of the skeleton was reported to be ~11.5 GPa, much higher than other body tissues.[22]

The PMM is assumed to be a material following neo-Hookean laws. However, there is limited information regarding the passive viscoelastic mechanical behavior of the PMM. Similar to skin, the same Poisson's ratio (0.49) was set for the PMM.[18] One of the major objectives in this study was to determine the material coefficients of the PMM for senior women. As mentioned above, the PMM can be divided into the clavicular and sternocostal portions. According to research,[23] the Young's modulus of the clavicular portion can be calculated as 0.206 MPa while that of the sternocostal portion is 0.136 MPa during abduction. By taking these values as the reference, we have tried a series of parameters to optimize the FEM of the PMM. In this work, Marc MSC (MSC, USA) was employed as finite element software to construct a comprehensive musculoskeletal model to simulate the biomechanical behavior of the PMM in the arm abduction from 30° to 90°.

Mesh creation. Mesh creation of the geometric model was accomplished by Apex. The 3D four-node tetrahedral solid element was used to discretize the model for the computational domain. The reason to select this element is that its quadratic displacement behavior is quite suitable for the modeling of the irregular shape deformation often found on the female breast and PMM. The landmarks A, B, C and O are described in the mesh (Figure 3(d)) and the detailed relative angle and distance are represented in the experimental result by using the motion capture system. Afterwards, the PMM and the whole torso were meshed into a large number of solid elements. The total number of solid elements was 32,085 for the body, 19,495 for the torso (excluding the breast), 6468 for the breast, 2714 for the skin, 1046 and 2362 for the clavicular and sternocostal portions, respectively.

Loading and boundary conditions. The refinement of loading and boundary conditions is extremely crucial for an accurate FEM. Because of the symmetric body structure, the simulation was performed using the half body for higher efficiency. As participants were required to wear a standard bra during 3D body scanning, gravity force was exerted as compensation on the breast by 10 increments. A rotation centre can be measured as the intersections by the extension of humerus under arm abduction from 30° to 90°. The central point of the end of the humerus was defined as the rotation center, which was (136.01, 39.7324, –10.5355).

Results and discussion

Women's perceptions on sports bras

A questionnaire was carried out for the women to inspire the design of sports bras. The one-way analysis of variance (ANOVA) results of the sensor women's attitude on different purchasing factors is shown in Figure 4. Among various purchasing factors, quality has a significant difference in the supporting percentage when compared with asthetics and recommendations, while function has a similar result to the quality. Statistically, a significant difference was also found between the comfort and recommendations. Hence, women seem rather objective when they decide to purchase a sports bra. In other words, a sports bra with better qualities and more functions will be highly appreciated in comparison to the beautiful and recommended bra.

In terms of the women's torso, underbust was considered as the most uncomfortable position for both daily and sports bras (Figure 5). It is concluded that a potentially fascinating sports bra can be designed with the structural materials to improve the quality and also the comfort of the underbust. Therefore, in this work, we aimed to create the FEM based on the women's torso to investigate breast deformation, which in turn facilitates the ergonomic design of the sports bra to motivate senior women to exercise.

Skin extension of breast and material properties determination

Skin extension of the breast. The deformation ratio of OA, OB and OC can be calculated by the following equation:

$$\lambda = \left(\frac{\Delta L' - \Delta L_0}{\Delta L_0} \right) \times 100\% \qquad (1)$$

where $\Delta L'$ presents the original displacement while $\Delta L'$ presents the deformed displacement.

The results derived from the motion capture system are shown in Figure 6 and Figure 7. It was suggested that the largest skin strain of the breast is OC. Six participants have a similar changing trend, in which the average value of the final changing ratio of OA was changed by 2.5%. Compared with OA, there is a relatively larger deviation of the breast deformation as the arm abducts by 60°. OC has a nearly four times larger deformation than OA. The difference between OA and OC may be due to the collagen fiber orientation of human skin. It was usually found in the longitudinal direction in the upper area of the breast. This result is important to design the sports bra by applying

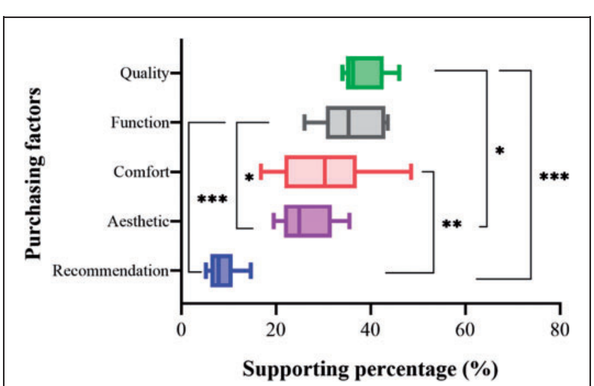

Figure 4. The factors for purchasing the sports bra; *, ** and *** indicate the significant difference: P < 0.05, P < 0.01 and P < 0.001, respectively.

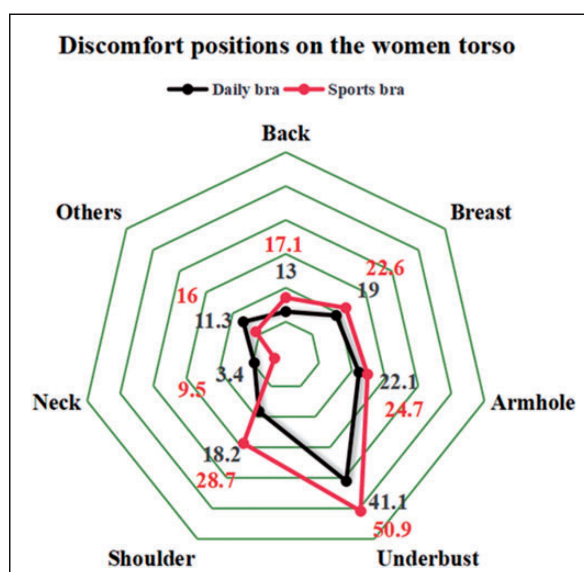

Figure 5. The rader chart on discomfort positions when women were wearing a sports bra.

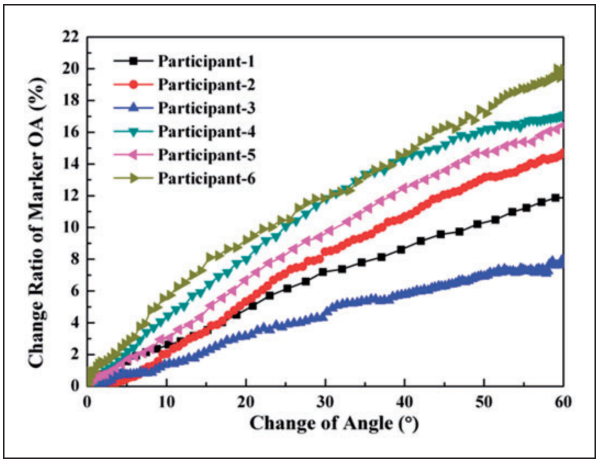

Figure 6. The distance changing ratio of OA as the arm rises from 30° to 90°.

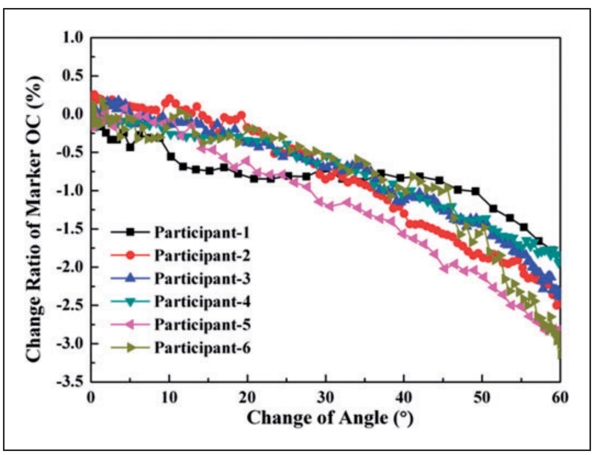

Figure 7. The distance changing ratio of OC as the arm rises from 30° to 90°.

more elastic materials to dissipate the large tension because of deformation in specific areas.

Material properties determination. As indicated by the motion capture results, the rough value of material coefficient for the sternocostal portion in the PMM was tuned from 0.06 to 2.0 with a compensation value of 0.5.

To validate the accuracy of the FEM by experiments, a method of root mean square error (RMSE) can be used:

$$\mathrm{RMSE} = 100\% \times \frac{1}{n}\sqrt{\sum_{i=1}^{n}(\Delta\mathrm{L}_{\mathrm{Exp},i} - \Delta\mathrm{L}_{\mathrm{FEM},i})} \quad (2)$$

Here, ΔL_{Exp} and ΔL_{FEM} are the deformation ratios obtained from the experiment and simulation, respectively; n is the number of data points generated during arm abduction. RMSE results reflect the data difference, and the lower the better. Eventually, optimized material coefficients can be determined by the minimum value of RMSE results.

In this work, one of the aims was to determine the biomechanical behavior of the PMM for the elderly by the analysis of theoretical and experimental data. To be specific, the Young's modulus of the designated clavicular and sternocostal portion can be calculated through the variable-controlling method. Given that the Young's modulus of the clavicular portion was determined as 0.206 MPa for the young,[18] it is reasonable to set relatively lower values of 0.2, 0.15, 0.1 and 0.05 MPa for the elderly. Correspondingly, a series of values for the sternocostal portion were validated in a range from 0.2 to 0.06 MPa (Table 3). As shown in Figure 8, the tendency is quite similar for 0.2, 0.15 and 0.05 MPa. As known from RMSE results, the minimum value of OA + OC was clearly found to be 0.1 MPa (clavicular portion) and 0.08 MPa (sternocostal portion), when the simulating data were closest to the experimental value. Notably, the minimum RMSE value was determined as 0.0849%, which was far lower than the acceptable limit (<0.5%). Therefore, it is confirmed that the FEM with specified material coefficients for different tissues can be successfully applied to predict breast deformation during arm abduction.

Figure 9 shows the comparison of FEM results and motion capture data to analyze the marked OA and OC during arm abduction. The simulating result was in line with the experimental data. Interestingly, the displacement change ratio of OC was positively related to the change of angle while an opposite trend was observed for OA. It indicated that the OA and OC were, respectively, under contraction and extension during arm abduction. It is noteworthy that the total displacement change of OC was almost four times higher than that of OA when abducting the arm from 30° to 90°.

FEM analysis

As the material parameters of the PMM were confirmed, loading conditions were investigated for the displacement of the PMM and breast. The contour band limitation of displacement was set as 0∼5 mm in order to detect clearly the characteristics of the breast displacement. As shown in Figure 10, a significant difference of displacement was observed for the breast with/without gravity. Figure 10(a) and (b) illustrated the differences between unloading and under gravity of the breast, while Figure 10(c) (PMM without gravity) and (d) (PMM under gravity) showed the gravity effects on the PMM. By the aid of sports bras, an

Table 3. The root mean square error (RMSE) results of the sternocostal portion of the pectoralis major muscle (PMM) in the finite element model (FEM)

		Sternocostal portion (MPa)							
Calvicular portion (MPa)	Marker	0.2 RMSE (%)	0.18	0.16	0.14	0.12	0.10	0.08	0.06
0.2	OA	0.0622	0.0662	0.0622	0.0619	0.0838	0.0615	0.0613	0.1188
	OC	0.1021	0.1155	0.1021	0.0996	0.1561	0.0937	0.0911	0.0736
0.15	OA	0.0628	0.0627	0.0625	0.0623	0.103	0.062	0.0615	0.1007
	OC	0.1061	0.1049	0.1025	0.1003	0.1994	0.0943	0.0918	0.1816
0.10	OA	0.0713	0.0703	0.0698	0.0624	0.0959	0.0621	0.0517	0.0885
	OC	0.1303	0.1271	0.1233	0.1006	0.1699	0.095	0.0482	0.0713
0.05	OA	0.0634	0.063	0.0629	0.0517	0.0712	0.0625	0.0632	0.0888
	OC	0.1077	0.1056	0.1035	0.0482	0.1211	0.0954	0.0954	0.1576

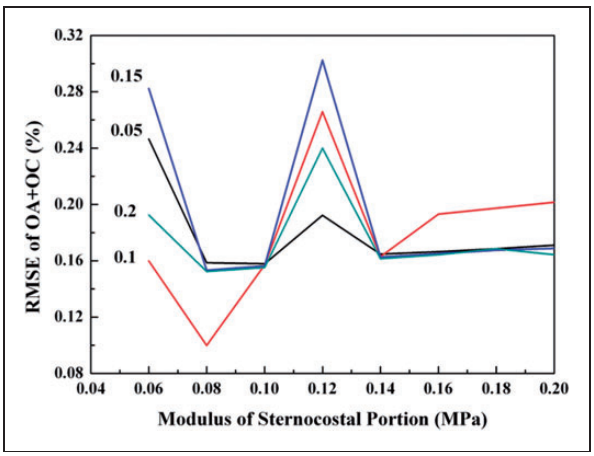

Figure 8. Validation of Young's modulus of clavicular and sternocostal portions within the pectoralis major muscle (PMM).

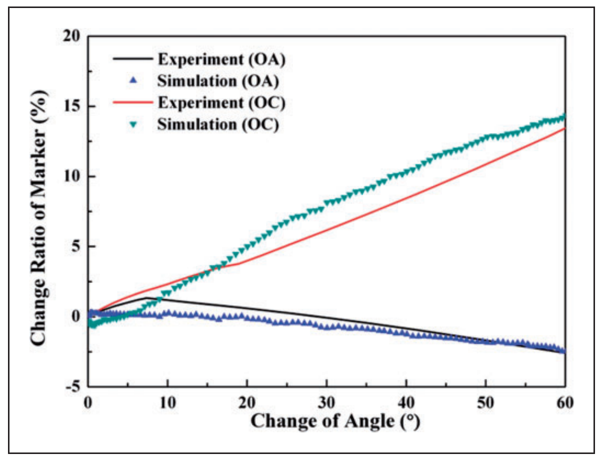

Figure 9. Comparison between the finite element modelling results and the motion capture experimental results.

average breast displacement of 3 mm was effectively eliminated, whereas no remarkable changes were observed for the PMM under distinct loading conditions. At the static stage, gravity-induced breast deformation is independent of the PMM but highly related with the support of skin and Cooper's ligament inside the breast.

To exploit further the relationship between the breast and the PMM, the arm abduction from 30° to 90° was conducted in the FEM (Figure 11). During the arm abduction process, both the breast and the PMM exhibited very similar trends in the displacement changes, which became more and more pronounced along with the arm abducting upwards. As expected, the breast and PPM were subjected to increasing displacement changes in the horizon direction, in which maximum changes were generated near the armpit. Although the displacement change of the breast is not so uniform as that of the PMM, their synergistic results have proved the aforementioned hypothesis that the breast deformation was largely affected by the contraction or extension of the PMM.

The displacement of the breast in x, y and z directions is shown in Figure 12. It could be seen that the largest displacement change occurs in the x direction, followed by the medium and smallest displacement change in the y and z directions, respectively. It suggested that the related breast deformation was mainly based on the two-dimensional plane and controlled by the PMM in the horizontal direction. This result may be interesting for the bra designer to improve bra stability through some specific support.

The results suggest that the upper outer area of the breast needs more support in the sports bra design to reduce the soft tissue deformation, while the inner upper area of the breast needs more space as the skin contracts during arm abduction. Fabrics with higher elasticity can be utilized here. Beside this, gore height can be designed higher to isolate the edges of two

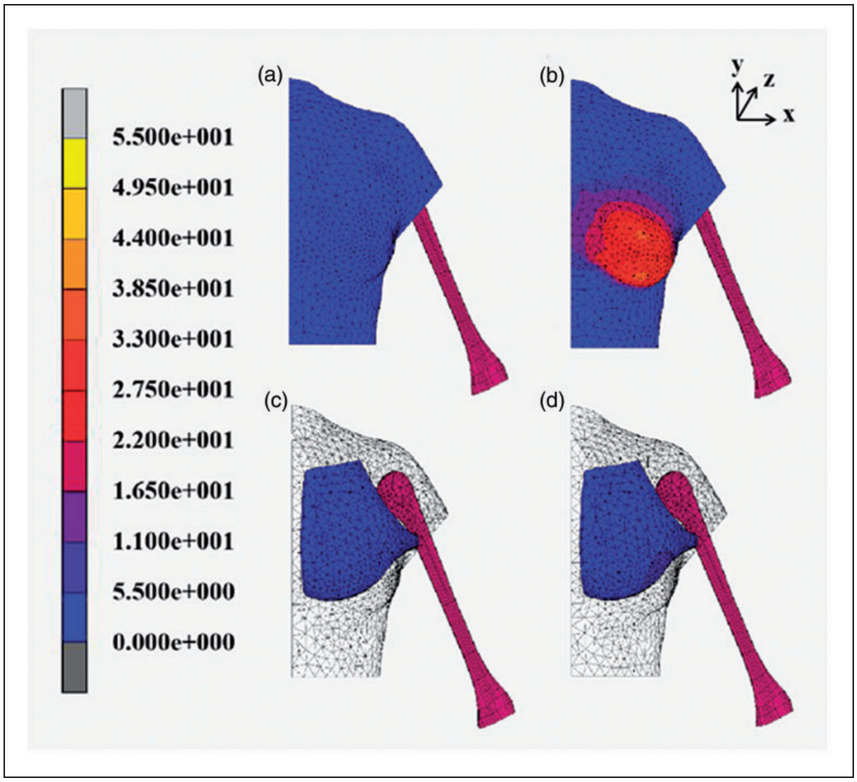

Figure 10. The comparison of the breast displacement and PMM displacement with and without the effect of gravity. (a) Breast without gravity; (b) breast with gravity; (c) PMM without gravity; (d) PMM with gravity.

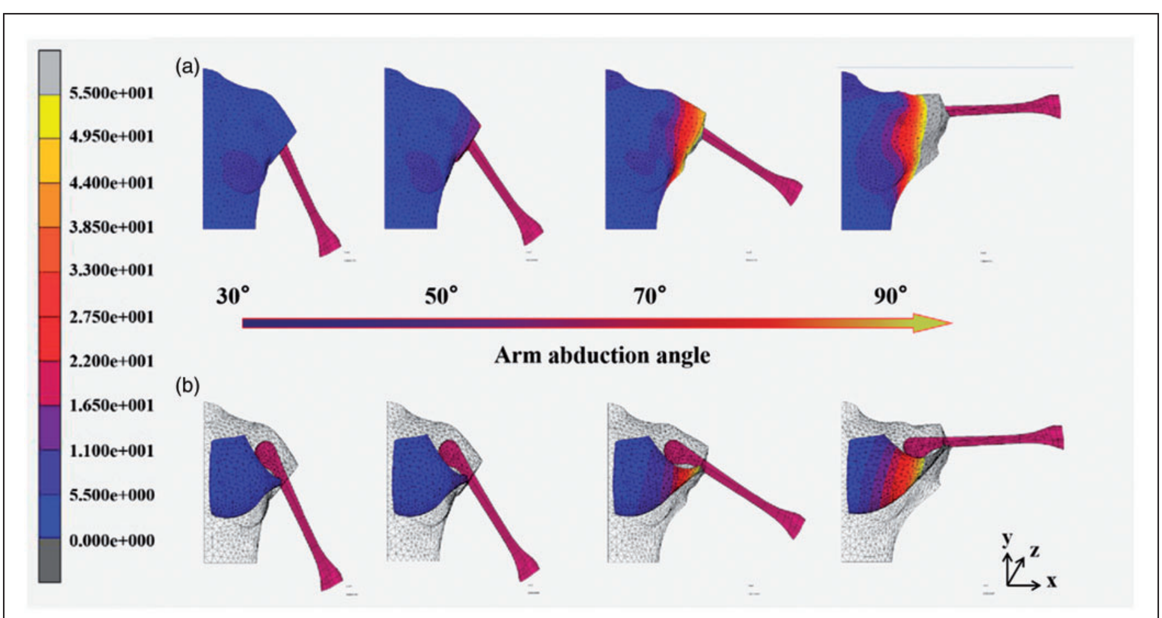

Figure 11. The simulation of total displacement of the breast during the abduction from 30° to 90°.

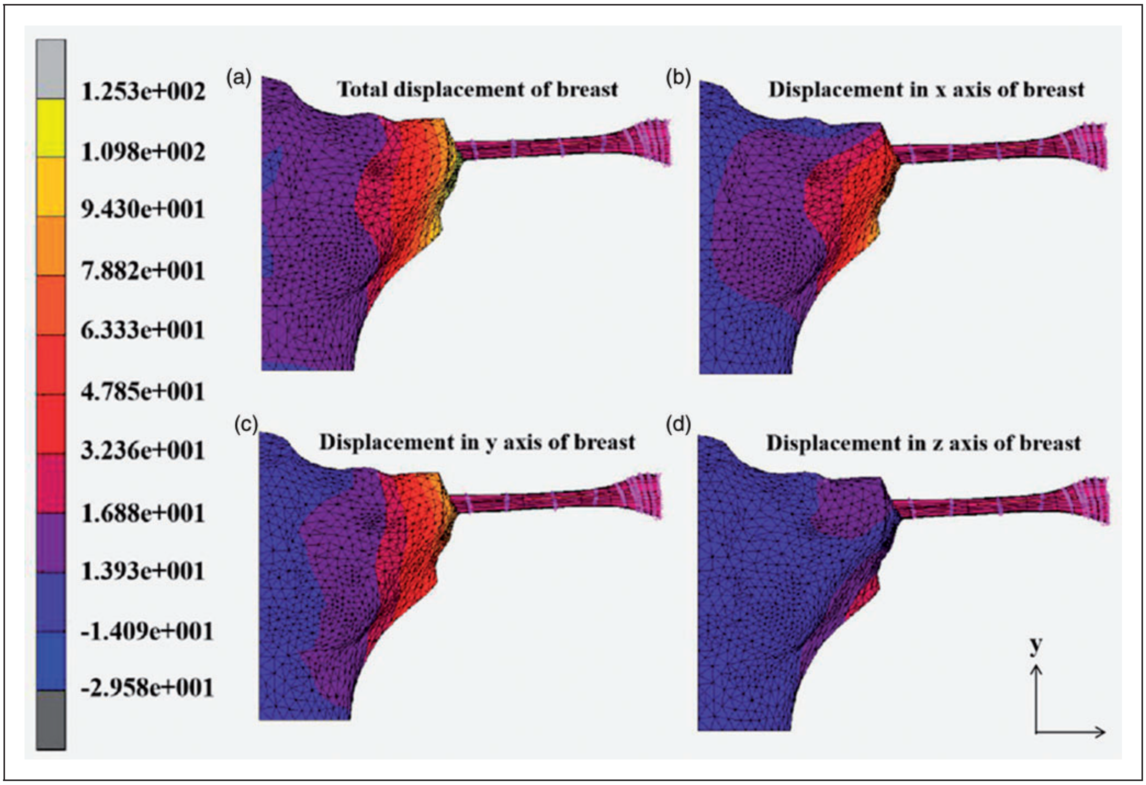

Figure 12. The total displacement of the breast and displacement in *x*, *y* and *z* directions.

breasts, but the gore width needs to be designed smaller to avoid skin contraction. What is more, the skin strain around the armhole area is so large that more elastic fabrics should be used.

Conclusions

A nonlinear multi-component FEM has been established to simulate the breast deformation of senior women during arm abduction. Through 3D body scanning, a torso including the breasts was taken as the geometric model to represent the whole musculoskeletal system. Thanks to agreement of the simulation data and experimental results, the Young's modulus of the PMM was finally determined with the clavicular and sternocostal portion of 0.1 MPa and 0.08 MPa, respectively. Notably, the RMSE is as low as 0.0849%, reflecting a reliable result to confirm material parameters. A synergistic effect was observed from the independent FEM between the breast and the PMM. In other words, the contraction and extension generated by the PMM made significant contributions to the final breast deformation, allowing the prediction of large breast motions for senior women simply based on the FEM of the PMM. It is worth noting that several contact bodies from the combinations of breast, torso, PMM and bones were involved in this FEM. Some large strain generated from soft tissues may cause certain difficulty in building the model. To elucidate further the benefits of wearing a sports bra, deformable sports bras can also be introduced into the FEM, investigating the interaction among breast, torso, PMM and bra. Besides this, by regulating the bra parameters, it is more straightforward to understand the bra performance in terms of support, pressure and motion control. Interestingly, the data derived from the FEM provided potential benefits in the ergonomic design of sports bras for senior women to provide sufficient protection and exercise motivation. Furthermore, this model laid a foundation for the elderly breasts' mechanics during arm abduction, which may be potentially extended to other sports activities.

Declaration of conflicting interests

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References

1. Howdon D and Rice N. Health care expenditures, age, proximity to death and morbidity: Implications for an ageing population. *J Health Econ* 2018; 57: 60–74.
2. Kydland F and Pretnar N. *The costs and benefits of caring: Aggregate burdens of an aging population* (No. 25498). National Bureau of Economic Research, Working Paper Series, Massachusetts, USA, 2019.
3. Landi F, et al. Impact of habitual physical activity and type of exercise on physical performance across ages in community-living people. *PLoS One* 2018; 13: e0191820.
4. McAuley E and Rudolph D. Physical activity, aging, and psychological well-being. *J Aging Phys Activity* 1995; 3: 67–96.
5. McGhee D. *Sports bra design and bra fit: Minimising exercise-induced breast discomfort*. Doctoral Thesis, University of Wollongong, Wollongong, Australia, 2009.
6. Han L, et al. Development of patient-specific biomechanical models for predicting large breast deformation. *Phys Med Biol* 2011; 57: 455.
7. Annaidh AN, et al. Characterization of the anisotropic mechanical properties of excised human skin. *J Mechan Behav Biomed Mater* 2012; 5: 139–148.
8. Norris M, et al. Do static and dynamic activities induce potentially damaging breast skin strain? *BMJ Open Sport Exerc Med* 2020; 6: e000770.
9. McGhee DE and Steele JR. Breast elevation and compression decrease exercise-induced breast discomfort. *Med Sci Sports Exerc* 2010; 42: 1333–1338.
10. Liang R, et al. Numerical simulation of nonlinear material behaviour: application to sports bra design. *Mater Design* 2019; 183: 108177.
11. Sun Y, et al. 3D Bra and human interactive modeling using finite element method for bra design. *Comput-aid Design* 2019; 114: 13–27.
12. Li Y, Zhang X and Yeung KW. A 3D biomechanical model for numerical simulation of dynamic mechanical interactions of bra and breast during wear. *Sen'i Gakkaishi* 2003; 59: 12–21.
13. Azar FS, Metaxas DN and Schnall MD. *A finite element model of the breast for predicting mechanical deformations during biopsy procedures*. Proceedings IEEE Workshop on Mathematical Methods in Biomedical Image Analysis. MMBIA-2000 (Cat. No. PR00737). IEEE 2000; 38–45.
14. Laan DV, et al. Chest wall thickness and decompression failure: a systematic review and meta-analysis comparing anatomic locations in needle thoracostomy. *Injury* 2016; 47: 797–804.
15. Coltman CE, Steele JR and McGhee DE. Effect of aging on breast skin thickness and elasticity: implications for breast support. *Skin Res Technol* 2017; 23: 303–311.
16. Page KA and Steele JR. Breast motion and sports brassiere design. *Sports Med* 1999; 27: 205–211.
17. Gefen A and Dilmoney B. Mechanics of the normal woman's breast. *Technol Health Care* 2007; 15: 259–271.
18. Ramião NG, et al. Biomechanical properties of breast tissue, a state-of-the-art review. *Biomechan Model Mechanobiol* 2016; 15: 1307–1323.
19. Dong SL, et al. Development of an adjustable model breast for mammographic dosimetry assessment in Taiwanese women. *Am J Roentgenol* 2011; 196: W476–W481.
20. Wolfe SW, Wickiewicz TL and Cavanaugh JT. Ruptures of the pectoralis major muscle: an anatomic and clinical analysis. *Am J Sports Med* 1992; 20: 587–593.
21. Butt U, et al. Pectoralis major ruptures: a review of current management. *J Shoulder Elbow Surg* 2015; 24: 655–662.
22. Hodgson JA, et al. Finite element modeling of passive material influence on the deformation and force output of skeletal muscle. *J Mechan Behav Biomed Mater* 2012; 9: 163–183.
23. Ward SR and Lieber RL. Density and hydration of fresh and fixed human skeletal muscle. *J Biomechan* 2005; 38: 2317–2320.